MotionTennis

Milestone M2
Requirements specification and Gate 1

Submission field	Value
Team name	MotionTennis
Milestone	M2 - requirements agreed / Gate 1
Report date	11 November 2026
Revision and status	R4 - audited baseline; historical chronology issue (Q-12)
Team approval date	26 October 2026 - confirmed by the team

TEAM REQUIREMENTS APPROVAL
The team has read and approved the requirements F-01-F-21 and NF-01-NF-15 in this report, including the explicitly recorded deferrals. Approval date: 26 October 2026. Historical compliance with the approval-before-architecture rule remains unresolved (Q-12).

Purpose

Define what Motion Tennis must do, why each requirement exists, and how it can be tested. This self-contained M2 baseline separates requirements from the architecture and detailed transaction budgets due at M3. [L-01 pp. 7-8]

Owners are sub-team roles only. The report includes no student names, personal identifiers, credentials, repository histories or screenshots. Numerical targets are project choices, not values assigned by the lecture. [NF-01-NF-04, NF-11]

01 / SYSTEM AND VERIFICATION

System in context

Motion Tennis is a motion-controlled tennis demonstrator for four humans playing 2v2 doubles. Its field of application is the study of distributed processing, communication latency, synchronization and data freshness through an interactive game. Players use wireless battery handhelds and active camera observations while virtual players move automatically. [F-01, F-03, F-13, F-16]

Physical loop: human arm/controller movement > IMU and camera observations > associated and calibrated racket/ball decision > accepted visual state and eligible electrical/haptic feedback > the player perceives the result and moves again. [F-02-F-08, F-20]

Scope and boundaries

Included requirement scope	Boundary or deferral
Four humans, stable teams, automatic virtual movement and physical racket/serve gestures. [F-16-F-20]	One-to-three-controller configurations are diagnostic workloads, not complete doubles matches. [F-01, F-16]
Camera/IMU fusion with calibrated constrained geometry. [F-03, F-13]	Unrestricted true 3D is deferred because monocular observation does not establish metric wrist depth. [F-13; D-02]
Explicit state query/response, ordered accepted results and semantic feedback. [F-04-F-09]	A technical fault is not a sporting miss, double fault or awarded point. One fail-stop loss requires safe degradation; continuous play is not guaranteed. [F-06, F-07, F-18, F-21]
At least five physical nodes, custom sensing/actuation and measurable deadlines. [NF-01, NF-12]	Target hardware proof, detailed stage budgets, global clock granularity and safety limits remain explicit later evidence or deferrals. [NF-09, NF-13-NF-15; Q-06, Q-10]

Verification and traceability

Every F-/NF- entry gives a behavior, reason and falsifiable test. Expected fixture outcomes must be declared before execution. A fixture pass establishes only that case; physical and performance acceptance require target runs. The tests in this report are specified, not reported as executed. [NF-04, NF-05, NF-10]

F-01-F-20 preserve established identifiers; F-21 makes the lecture fault hypothesis explicit. Earlier N-01-N-11 identifiers map one-to-one to NF-01-NF-11. NF-12-NF-15 address course and checklist obligations. Ranges are inclusive.

L-01 identifies the lecture slides; C-01 identifies the checklist referenced by slide 12. Section 04 separates lecture methodology from project numerical choices. Section 05 maps all 13 DRS and 8 prerequisite embedded-system checklist rows. [NF-01-NF-04, NF-11-NF-15]

02 / FUNCTIONAL REQUIREMENTS / 1 OF 4

Functional requirements F-01 to F-05

Behavior and observable responsibilities; software structure, transport and deployment are deferred to system design.

ID	Requirement and reason	Falsifiable acceptance criterion
F-01	**Support one to four active wireless, battery-powered handheld controllers.** **Reason:** Permit incremental testing and four-person doubles.	Run with each controller count from one to four. Every admitted controller has a distinct session identity and separately reconciled inputs and outcomes. A full match still requires four humans under F-16.
F-02	**Create one identified transaction at each regular IMU sampling boundary. Project target: 200 Hz per controller (5 ms nominal interval).** **Reason:** Preserve the offered workload and expose sample loss.	Verify configuration/readback for 200 Hz and measure cadence during NF-04 runs: measured frequency must agree with 200 Hz within declared instrument/clock uncertainty. Report interval/jitter distributions; the jitter limit is deferred Q-10. Each boundary has original t0 and a unique key; count all admissions/rejections. Markers do not gate/add transactions. [L-01 pp. 3, 11; C-01 DRS-09; rate is a project choice.]
F-03	**Use active camera body/joint observations together with IMU observations to derive racket gestures.** **Reason:** Body-relative position and orientation need complementary observations.	Controlled fixtures vary each source independently and demonstrate its effect on eligible gesture results. Missing, invalid or unsuitable required observations cause an explicit technical outcome rather than a fabricated hit or sporting miss.
F-04	**Obtain an identified ball-state request and matching response for each valid physics transaction.** **Reason:** Make each calculation traceable to the state it uses.	Each valid transaction has a logical query, matching response and snapshot version or tick in its trace. A shared exchange preserves every logical transaction key. Invalid input may be rejected before querying; an unsolicited old snapshot cannot substitute.
F-05	**Calculate hit/miss and resulting ball motion from eligible, associated inputs.** **Reason:** Translate physical gestures into consistent tennis interaction.	Deterministic hit, no-contact, edge-contact and repeated-contact fixtures produce the declared ball result. A contact produces at most one hit event. Invalid input never becomes a normal miss.

Verification owners: controller, camera tracking, physics, game presentation and debug support; integration support checks the whole physical loop. [F-01-F-21; NF-05]

02 / FUNCTIONAL REQUIREMENTS / 2 OF 4

Functional requirements F-06 to F-10

Behavior and observable responsibilities; software structure, transport and deployment are deferred to system design.

ID	Requirement and reason	Falsifiable acceptance criterion
F-06	**Account for every generated transaction with a correlated result or visible local failure.** **Reason:** Prevent lost or failed work from disappearing.	At the reporting cutoff, each unique transaction has exactly one terminal outcome or an explicit pending status. Drop, error, overload and timeout counts remain separate. Duplicate or late responses cannot create a second completion.
F-07	**Distinguish a hit, a completed missed swing and a technical fault in feedback.** **Reason:** Help players interpret valid play and failures.	Hit and completed valid missed-swing fixtures yield distinguishable configured cues. A technical fault has an explicit fault indication. Ordinary no-contact samples do not fabricate haptic cues. Only eligible, timely results may activate gameplay outputs.
F-08	**Store and present one versioned, accepted match state, with one authority calculating its changes.** **Reason:** Keep ball, avatars, serving and score mutually consistent.	The calculation authority supplies match changes; the committed-state role stores accepted versions; presentation consumes those versions. Ball, avatar, serve and score changes appear atomically. Rejected changes are not displayed as accepted play.
F-09	**Resolve concurrent inputs and continuous motion into one ordered match history.** **Reason:** Avoid conflicting ball histories and duplicate points.	Simultaneous-input, replay, duplicate-update and no-controller-input fixtures preserve one ordered committed history. Ball motion continues between swings. No duplicate state transition or point is accepted, including after a reconnect.
F-10	**Preserve original source identities and timestamps through processing.** **Reason:** Enable causal reconstruction and defensible timing.	A trace audit recovers the selected IMU sample and camera frame/person, original capture times, receiver arrival, processing boundaries and logical simulation time. Each time retains its clock/session identity; arrival time never overwrites capture time.

Verification owners: controller, camera tracking, physics, game presentation and debug support; integration support checks the whole physical loop. [F-01-F-21; NF-05]

02 / FUNCTIONAL REQUIREMENTS / 3 OF 4

Functional requirements F-11 to F-15

Behavior and observable responsibilities; software structure, transport and deployment are deferred to system design.

ID	Requirement and reason	Falsifiable acceptance criterion
F-11	**Handle connection, player association, calibration, loss and restart explicitly.** **Reason:** Prevent wrong-player actions and unsafe stale work.	Disconnect, controller restart, tracking loss and invalid-calibration tests stop affected gameplay decisions until explicit readiness is restored. Work from an old session cannot activate a new session. Ambiguity never causes automatic reassignment to a nearby person.
F-12	**Provide diagnostic inspection of complete transactions and failures.** **Reason:** Make the distributed physical loop observable.	Select one success and each injected failure class. The diagnostic view reconstructs input, pose selection, ball-state query/response, calculation, accepted/rejected state change and feedback where reached. Missing records and unresolved links remain visible; they are never invented.
F-13	**Represent monocular gameplay geometry as calibrated 2D/2.5D estimates.** **Reason:** Avoid claiming unrestricted metric depth from one RGB view.	Published observations identify the constrained geometry, calibration and validity. Out-of-plane or invalid observations are rejected or explicitly marked unusable. No estimated free-space depth is presented as a verified measurement.
F-14	**Provide an independent configurable hardware acceleration-threshold marker for diagnostics.** **Reason:** Observe movement candidates without changing gameplay load.	Markers have their own identity and original trigger time, with an explicit resolved or unresolved link to regular transactions. Marker-enabled and marker-disabled trials preserve regular sampling, payload meaning and processing policy; a marker alone cannot create a hit or serve.
F-15	**Provide versioned reference-clock mapping with offset/drift correction and visible uncertainty.** **Reason:** Qualify cross-device alignment and timing claims.	The diagnostic view retains raw source time, source/reference session, mapping version and numeric uncertainty bounds. Missing, expired or invalid maps are visibly unusable for bounded cross-source claims. Local same-clock intervals remain available. NF-11 supplies candidate limits.

Verification owners: controller, camera tracking, physics, game presentation and debug support; integration support checks the whole physical loop. [F-01-F-21; NF-05]

02 / FUNCTIONAL REQUIREMENTS / 4 OF 4

Functional requirements F-16 to F-21

Behavior and observable responsibilities; software structure, transport and deployment are deferred to system design.

ID	Requirement and reason	Falsifiable acceptance criterion
F-16	**Provide four-human 2v2 doubles with automatic virtual-player locomotion.** **Reason:** Let humans supply racket gestures without chasing a virtual ball physically.	A full-match trial has four confirmed humans in two teams. Virtual avatars move according to the match model; assigned humans control racket and serve gestures. One-to-three-controller trials are diagnostic and do not silently insert computer players.
F-17	**Let only the assigned server initiate a controller-lift toss and overhead serving attempt.** **Reason:** Provide the agreed motion-based serving interaction.	Versioned fixtures distinguish another player's gesture, a cancelled toss, a legal serve, an attempted strike that misses, a first/second fault and a let. An acceleration marker alone never authorizes the serve. Recognition thresholds remain Q-04.
F-18	**Apply ordinary tennis scoring and explicit sporting-fault distinctions.** **Reason:** Make match outcomes understandable and repeatable.	Fixtures cover love/15/30/40, deuce/advantage, service and receiver rotation, first/second service fault, service let, line-in, net/out and second bounce. Cancelled toss is distinct from attempted miss. Technical failure awards no point. Match length/tie-break configuration remains Q-05.
F-19	**Show the same match from elevated rear views for both teams.** **Reason:** Give each team a readable view from its own court end.	Both labeled views consume the same committed version, ball and score. After end changes, views follow their teams without changing team identities, handedness or collision geometry. Screen arrangement remains Q-05.
F-20	**Require explicit controller/person/hand/team association and four-ready startup.** **Reason:** Prevent incorrect player attribution.	An operator confirms the four unique controller/person bindings and stable A1/A2/B1/B2 team slots. Duplicate or ambiguous bindings and stale installation acknowledgements are rejected. Full doubles cannot start until all four valid bindings acknowledge readiness.
F-21	**Tolerate one fail-stop node loss by entering a safe degraded state and explicitly controlling reintegration.** **Reason:** Prevent a failed participant from causing wrong-player output, fabricated points or inconsistent recovery.	Hypothesis: k = 1 non-malicious fail-stop node/link; continuous play is not required. Each injected loss pauses affected play, prevents unconfirmed cues/points and leaves surviving outputs safe. Detect partitions/cliques; separated groups cannot score independently. Rejoin validates current state, association/calibration and readiness; link recovery may retain its session, restart requires a fresh boot/session. Record specified beyond-k injections externally; no guarantee outside k. [L-01 p. 4; C-01 DRS-06, 08, 13]

Verification owners: controller, camera tracking, physics, game presentation and debug support; integration support checks the whole physical loop. [F-01-F-21; NF-05]

03 / NONFUNCTIONAL REQUIREMENTS / 1 OF 3

Nonfunctional requirements NF-01 to NF-05

Numerical targets are project choices requiring target validation. Lecture and checklist references identify the governing method; open limits have explicit deferrals.

ID	Requirement and reason	Falsifiable acceptance criterion
NF-01	**Bound and measure complete controller response and eligible electrical feedback.** **Reason:** Keep motion interaction responsive while rejecting expired work.	Project targets: response p99 <= 30 ms; eligible electrical feedback p99 <= 40 ms. At local 50 ms expiry, retire each incomplete transaction as one counted timeout and suppress new expired gameplay actuation. A late result cannot create another completion. Verify receipt, expiry and output on each controller under NF-04. Values are project choices. [L-01 pp. 3-5, 11; C-01 DRS-05]
NF-02	**Reject sensor data that exceeds declared age, alignment, prediction or uncertainty limits.** **Reason:** Fresh packet arrival does not establish fresh or compatible sensing.	Project upper bounds: IMU age 30 ms; pose age 80 ms; plane-position prediction 50 ms; historical IMU/pose alignment 25 ms; camera timestamp uncertainty 8 ms. Include uncertainty and recheck after state waiting. Boundary and exceeded-bound fixtures must pass or reject explicitly. Reading-task phase sensitivity is analyzed under NF-14. Values are project choices. [L-01 p. 3; C-01 DRS-03]
NF-03	**Bound queues, retained histories and scheduled workload.** **Reason:** Prevent overload from becoming uncontrolled backlog.	Project limits: four queued inputs/controller, eight inputs/batch, at most two/controller and one batch in flight. Release/step 5 ms; baseline batch-service p99 <= 4 ms. Catch up at most four steps; when lag exceeds 20 ms, pause/degrade rather than extend catch-up. Above-peak tests count rejections and missed releases, preserve queue bounds and show recoverable throughput without thrashing. Values are project choices. [L-01 pp. 3, 11; C-01 DRS-09]
NF-04	**Reconcile all attempts and report complete performance evidence.** **Reason:** Successful-only results conceal failures and deadline misses.	Project experiment: three 10 min baseline runs for each controller count 1-4 after separately recorded warmup. Require <= 1% late-or-failed attempts, NF-01 targets, no lost acceptance logs and zero observed wrong associations, duplicate commits or stale actuation. Report min, mean, median, p95, p99, observed maximum and failures with denominators; timeouts remain censored. Duration is a project choice. [L-01 pp. 5-6, 11]
NF-05	**Make builds, configurations and experiments reproducible.** **Reason:** Allow another sub-team to reproduce the same evidence.	An independent sub-team reproduces a declared functional test using a frozen manifest of device revisions, dependency versions, model/schema/profile checksums, calibration, operating environment, workload and startup steps. Missing manifest items fail reproducibility review.

03 / NONFUNCTIONAL REQUIREMENTS / 2 OF 3

Nonfunctional requirements NF-06 to NF-10

Numerical targets are project choices requiring target validation. Lecture and checklist references identify the governing method; open limits have explicit deferrals.

ID	Requirement and reason	Falsifiable acceptance criterion
NF-06	**Keep diagnostics off the synchronous gameplay path and measure observation overhead.** **Reason:** Logging must not silently change the behavior being measured.	Collector disconnection and full trace-buffer injection do not block gameplay processing. Every lost diagnostic record is counted. Matched logging-on/off trials report timing differences and losses. A quantitative acceptable-overhead budget is open in Q-08; none is invented here.
NF-07	**Maintain compatible interfaces and preserve declared properties during integration.** **Reason:** Prevent independently developed parts from interpreting data differently or losing verified behavior when combined.	Producers/consumers pass shared valid, malformed, identity, session, unit and maximum-size fixtures; unsupported versions fail explicitly. For each integration step, name the identity, atomicity and timing properties to preserve and rerun their criteria before and after integration. Any regression fails integration. [C-01 DRS-10]
NF-08	**Use English for newly authored project documentation and interfaces.** **Reason:** Keep shared work consistent across sub-teams.	A review finds English report content, UI labels, project-defined identifiers, comments and examples. Required programming syntax, proper names and unchanged legal attribution are preserved outside this anonymous report.
NF-09	**Keep controller power and feedback behavior within reviewed safe limits.** **Reason:** Battery handhelds and actuators must remain safe during faults.	Outputs start off and fail safe during reset, disconnect, malformed/duplicate/expired feedback and low-battery tests. Verify actual output effect, including a blocked-actuator test, rather than infer effect from a command. Electrical drive must stay within reviewed ratings. Pulse cap, endurance and detection thresholds are deferred Q-06 pending parts. No certified safety function is claimed. [C-01 DRS-12; EMB-02, 04]
NF-10	**State the real-time classification and measurement limits accurately.** **Reason:** Avoid turning a finite experiment into an unsupported guarantee.	Every performance conclusion distinguishes statistical service targets, stale-work rejection, electrical output, mechanical vibration and display light. No observed maximum is called a proven worst-case bound; no hard end-to-end guarantee is claimed without supporting worst-case evidence.

03 / NONFUNCTIONAL REQUIREMENTS / 3 OF 3

Nonfunctional requirements NF-11 to NF-15

Numerical targets are project choices requiring target validation. Lecture and checklist references identify the governing method; open limits have explicit deferrals.

ID	Requirement and reason	Falsifiable acceptance criterion
NF-11	**Qualify every cross-device comparison by mapping validity, drift and uncertainty.** **Reason:** Device offsets and drift can invalidate apparent duration and ordering.	Project policy: refresh mappings every 1 s, expire after 5 s without valid refresh and accept at most 2 ms mapping uncertainty per source. Tests cover offset/drift, asymmetric exchange, expiry and source/reference restart. Invalid maps suppress bounded cross-source claims while preserving raw records and same-clock durations. Values are project choices; they do not prove global precision. [L-01 pp. 10-11; C-01 DRS-01, 02]
NF-12	**Use at least five physical networked computing nodes and custom sensor/actuator hardware in the physical loop.** **Reason:** Meet the lecture exercise scope for a distributed real-time system.	The acceptance inventory identifies at least five independently executing physical computing nodes and demonstrates custom sensing and actuation in the loop. Multiple software processes on one physical computer count as one node. No allocation or topology is selected here. [L-01 pp. 3-4]
NF-13	**Establish measured global clock precision and a valid granularity for distributed timestamp reasoning.** **Reason:** A small local timestamp quantum does not establish a precise shared time base.	Define global precision pi and granularity g, both in ms. Target tests must establish g > pi; cross-node comparisons respect the checklist reasonableness condition and the +/-2g measurement limit. Display unresolved order when uncertainty overlaps. Exact pi/g values are deferred to Q-10 because target clock evidence is absent. [C-01 DRS-01, 02; L-01 pp. 10-11]
NF-14	**Provide a target-based timing-feasibility case for each transaction before claiming bounded real-time behavior.** **Reason:** Percentile targets alone cannot establish worst-case deadline compliance.	For every transaction, record its owner, input/action/transmit/receive/output boundaries, WCET, jitter and delay assumptions. A claimed bound passes only if justified worst-case stage totals plus uncertainties are <= its deadline and scheduling/reading-phase analysis passes. Define permanence/action delay and include delayed hidden channels. M3 analysis is deferred in Q-10; no hard bound is claimed. All times use explicit units. [L-01 pp. 3, 5, 7-8, 11; C-01 DRS-03-05, EMB-01]
NF-15	**Validate constrained resources, local bus behavior and power-fault recovery on the selected hardware.** **Reason:** A functioning algorithm is insufficient if target resources, communication or supply limits fail.	Before target acceptance, record RAM/flash/CPU/stack and power limits. High-water and above-peak tests must stay within those limits; bus-contention and bus-error injections must yield explicit outcomes. Brown-out/reset must restore safe outputs and require F-21 readiness. Exact hardware limits, sleep policy and watchdog strategy are deferred in Q-06 until parts are fixed. [C-01 EMB-03, 04, 07, 08; L-01 pp. 10-11]

04 / TIMING AND LECTURE PROVENANCE

Timing definitions and sources

t0 is the regular controller IMU sampling boundary; tr is complete receipt of a matching application response, retained only after validation; t1 is the declared eligible electrical LED/motor activation. Application response is tr - t0; electrical-feedback time is t1 - t0. Each interval uses one controller clock. Separate LED and motor onsets are retained if different. [F-02, F-07, F-10; NF-01]

The intervals include queues, pose waiting, the required ball-state request/response, calculation and acceptance waiting. They do not automatically include pre-t0 sensing, mechanical vibration or display photons. Noneligible inputs have no invented t1; cross-device durations use uncertainty, never half-RTT assumptions. [F-04, F-10, F-15; NF-01, NF-10, NF-11]

ID	Project target, with unit	Lecture / checklist basis
F-02	200 Hz/controller; 5 ms interval. Derived four-controller load: 800 transactions/s.	L-01 pp. 3, 11; C-01 DRS-09
NF-01	Response p99 <= 30 ms; eligible electrical p99 <= 40 ms; local expiry 50 ms.	L-01 pp. 3-5, 11; C-01 DRS-05
NF-02	IMU age 30 ms; pose age 80 ms; prediction 50 ms; alignment 25 ms; camera-time uncertainty 8 ms.	L-01 p. 3; C-01 DRS-03
NF-03	Release/step 5 ms; baseline batch-service p99 <= 4 ms; pause/degrade if simulation lag exceeds 20 ms.	L-01 pp. 3, 11; C-01 DRS-09
NF-04	Three runs of 10 min for each controller count 1-4; separately recorded warmup.	L-01 pp. 5-6, 11 (method only)
NF-11	Mapping refresh 1 s; expiry after 5 s; uncertainty <= 2 ms/source.	L-01 pp. 10-11; C-01 DRS-01, 02

Lecture source L-01

Distributed Real-Time Systems - Exercise, ws26, 23 October 2026, FH Hagenberg; supplied 12-slide deck. PDF page numbers equal printed slide numbers. Pages 3-4 require a physical loop, measurable deadlines and fault tolerance; p. 5 uses monotonic timing and maxima for budgets; p. 7 gives role/transaction ownership; p. 8 places detailed budgets at M3; pp. 10-11 require timestamping and target measurements. Page 12 identifies C-01.

Magnitude provenance and limits

The lecture supplies no MotionTennis rate or millisecond target above. Those magnitudes are project engineering choices. Past-year RTT numbers and warm-up requirements on pp. 5-6 are not system deadlines. p99 is a statistical target; report maxima, failures and censored timeouts as well. Hard-bound feasibility requires NF-14; isolated percentiles cannot be added to establish an end-to-end bound. [NF-01, NF-04, NF-10, NF-14]

B-01: M2 submission brief supplied by the team: required report content, anonymity, file naming and Gate 1 before architecture is written.

05 / LECTURE CHECKLIST / DISTRIBUTED

DRS checklist reconciliation

C-01: Requirement Checklists, version 4, build package 0.3.1, public course source referenced by L-01 p. 12; accessed 11 November 2026. The stable link fixes the reviewed version. The Moodle ZIP was not supplied. Row labels below are concise paraphrases; the DRS section also requires the embedded checklist on the next page.

Row / topic	Requirement or explicit deferral	Verification / owner
DRS-01 Coordinated nodes	F-15; NF-11, NF-13: reference time, measured precision pi and granularity g > pi.	Synchronization tests. Exact pi/g deferred Q-10 pending target evidence; integration support.
DRS-02 Cross-node timestamps	F-10, F-15; NF-13: reasonableness and +/-2g limits; raw times and uncertainty preserved.	Controlled offset/drift and timestamp-order tests; integration support.
DRS-03 Physical observations	F-03, F-13; NF-02, NF-14: point-of-use age and reading-task phase sensitivity.	Age-bound tests. Scheduling/phase proof deferred Q-10 to M3; physics.
DRS-04 Action messages	F-09, F-21; NF-01, NF-14: causality, expiry, permanence/action delay and hidden channels.	Delayed/reordered-channel test. Exact action-delay analysis deferred Q-10 to M3; system design.
DRS-05 Distributed transaction	F-04-F-08; NF-01, NF-14: whole-loop deadline and stage-wise worst-case case.	Target stage timings and worst-case sum <= deadline. Budget allocation deferred Q-10 to M3; system design.
DRS-06 Node failure	F-21: k = 1 fail-stop/link loss; safe degradation, no uninterrupted-play or beyond-k guarantee.	Per-node and specified beyond-hypothesis injections with external outcome records. Redundant service deferred Q-11: outside initial scope; integration support.
DRS-07 Replica voting	Deferred Q-11: no voting replicas in initial scope, so exact/inexact voting is not used.	Any future replication must define determinism, voting and redundancy before release; system design.
DRS-08 Join / leave	F-11, F-20, F-21: detect inconsistent membership/cliques and partitions; affected play pauses without independent scoring.	Partition, membership-conflict and rejoin tests; integration support.
DRS-09 Peak load	F-02; NF-03, NF-15: declared node/channel load and bounded resources.	Above-peak load/throughput tests without thrashing; physics with integration support.
DRS-10 Integration properties	NF-07: preserve stated identity, state and timing properties during integration.	Compare the same acceptance properties in isolation and after integration; integration support.
DRS-11 Cluster gateway	Deferred Q-11: no cross-cluster gateway requirement in initial scope.	If introduced, define both flow-control modes, buffering/skipping and overflow test; system design.
DRS-12 Actuator safety	NF-09; teaching demonstrator only. No safety-certified function is claimed.	Observe the physical effect and blocked-actuator behavior. Exact limits deferred Q-06 pending parts; controller.
DRS-13 Reintegration	F-11, F-20, F-21: safe ground state, validated session/state and readiness; a restart needs a fresh boot/session.	Restart and link-recovery tests; reject stale history/cues. Minimal recovery state deferred Q-10 to M3; integration support.

05 / LECTURE CHECKLIST / EMBEDDED

Prerequisite embedded checklist

C-01 makes these eight embedded-system rows additional prerequisites for the DRS field. Each row is covered by a requirement or an explicit scope/evidence deferral. The external checklist is reference material; its tool-specific workflow is not adopted. [NF-09, NF-14-NF-15]

Row / topic	Requirement or explicit deferral	Verification / owner
EMB-01 Real-time behavior	NF-01, NF-10, NF-14: deadlines, WCET, jitter and scheduling evidence.	Target timing analysis; detailed proof deferred Q-10 to M3; system design.
EMB-02 Safety function	Deferred Q-11: a teaching/game demonstrator is in scope; a regulated or certified safety function is outside scope.	Review scope before any safety-critical deployment; requirements and product owner. Ordinary electrical safety remains NF-09.
EMB-03 Resource limits	NF-03, NF-15: RAM, flash, CPU and stack budgets.	Static/resource review and target high-water tests. Exact budgets deferred Q-06 pending actual hardware; component owners.
EMB-04 Battery / power	F-01; NF-09, NF-15: power budget, brown-out recovery and declared sleep policy.	Power and brown-out tests. Battery duration and sleep decisions deferred Q-06 pending parts/current measurements; controller.
EMB-05 Field update	Deferred Q-11: supervised semester prototype; no unattended field-update service required.	Before field deployment, require authenticated update/rollback and recovery tests; controller.
EMB-06 Nonvolatile wear	Deferred Q-11 until the persistent-write inventory is fixed; no high-rate gameplay logging to flash is required.	Any persistent-writing design must budget endurance/wear and test it before target release; controller.
EMB-07 Watchdog	NF-15; exact watchdog presence/kick/recovery policy deferred Q-06 because board/firmware setup is not fixed.	If present, stall/kick/reset injections must restore NF-09 safe output and F-21 readiness; controller.
EMB-08 Local buses	NF-07, NF-15: declared sensor-bus configuration, contention and explicit bus errors.	Target stress/error tests and recorded bus/driver versions; controller.

Deferral discipline

A deferral states what is not yet fixed, why, who owns closure and the evidence needed. A deferred proof is not a passed proof. The approved baseline permits later design choices only within these requirements; changing a target or fault hypothesis requires a recorded team decision. [NF-05, NF-07, NF-10, NF-14; Q-06, Q-10, Q-11]

06 / DECISIONS

Recorded scope decisions

The team approval covers this requirement baseline and its explicit deferrals. The choices below define scope and observable behavior; they do not select component deployment or communication design. [F-01-F-21; NF-01-NF-15]

Decision	Choice and reason	Trace
D-01	Four-human doubles with automatic avatar locomotion: players supply gestures without physically chasing a virtual ball.	F-01, F-03, F-16, F-20
D-02	Webcam-first constrained 2.5D: unrestricted metric depth cannot be established from one RGB view, so true 3D is deferred.	F-03, F-13
D-03	Assigned-server lift/toss and overhead serving: provides a recognizable physical interaction and protects turn ownership.	F-17, F-20
D-04	Tennis scoring separates sporting faults from technical interruptions: tracking/network failure must not invent a point.	F-06, F-07, F-18
D-05	Two elevated rear team views from the same accepted state: both teams need a readable, consistent match.	F-08, F-19
D-06	Explicit confirmed player association and readiness: discovery order or image position alone cannot establish identity.	F-11, F-20
D-07	Independent diagnostic markers and raw timestamps: observation must not change regular workload or hide timing uncertainty.	F-02, F-10, F-14, F-15; NF-06, NF-11, NF-13
D-08	Whole-loop measurement and complete accounting: pings and successful-only percentiles cannot establish physical-loop performance.	F-04, F-06; NF-01, NF-04, NF-10, NF-14
D-09	One fail-stop loss causes safe degradation and controlled rejoin: preserve valid state and player safety rather than promise uninterrupted play.	F-21; NF-09
D-10	At least five physical nodes and target-based evidence: follow the course scope and distinguish specified targets from measured guarantees.	NF-12, NF-14, NF-15

Team approval does not establish target performance. Match format, gesture tuning, hardware safety limits, data policy and detailed M3 feasibility evidence remain role-owned open items. [Q-04-Q-11]

07 / OPEN QUESTIONS

Questions and role owners

The team name is confirmed as MotionTennis and approval is recorded for 26 October 2026. This report was prepared and audited on 11 November 2026. Remaining questions have role owners. [F-01-F-21; NF-01-NF-15]

ID / owner	Question and evidence needed to close	Trace
Q-04 **Camera tracking with physics**	Fix physical playing area, four-person coverage, geometric/gesture accuracy, calibration thresholds, serve/swing thresholds and assistance. Close with declared physical or labeled-fixture evidence.	F-03, F-05, F-13, F-16, F-17; NF-02
Q-05 **Game presentation with physics**	Fix match length, tie-break/end-change configuration and screen arrangement. Close with versioned scoring fixtures and a two-view review.	F-17-F-19
Q-06 **Controller and component owners**	Pin board/sensor/actuator/power parts and resource budgets; specify safe drive/pulse limits, low battery, endurance, sleep and watchdog policy. Close with electrical, power, resource and bus tests.	F-01, F-07, F-14; NF-09, NF-15
Q-07 **Integration support with physics**	Fix physical-onset boundaries, instruments and accuracy. Validate response targets, sensor ages and clock uncertainty on target; revise only by recorded team decision.	F-10, F-15; NF-01, NF-02, NF-10, NF-11
Q-08 **Debug support with integration support**	Specify acceptable diagnostic overhead, full experiment load/warmup, trace retention and viewer acceptance. Close with a declared protocol, matched trials and complete loss accounting.	F-12, F-14; NF-04-NF-06
Q-09 **Integration support with camera tracking**	Define deployment trust boundary, source authorization, imagery retention, data minimization/access and deletion. These policies are not established by the current baseline.	F-03, F-11, F-12; NF-05, NF-07
Q-10 **System design with integration support**	At M3, fix sampling jitter limits and transaction owners; derive stage budgets, WCET/jitter, permanence/action delay, reading-phase analysis, measured pi/g and minimal reintegration state. Justify bounded claims or report infeasibility.	F-02, F-21; NF-07, NF-13, NF-14
Q-11 **Requirements and product owner**	Retain or revise deferred scope: voting/fail-operational redundancy, cluster gateways, regulated safety functions, field OTA/rollback and nonvolatile wear. Reopen before introducing such features.	F-21; NF-09, NF-15; C-01 DRS-06, 07, 11 and EMB-02, 05, 06
Q-12 **Requirements and product owner**	The team confirms approval on 26 October 2026; existing architecture records are dated 24 October 2026. Resolve this historical sequence with the instructor or establish that the earlier records are inaccurately dated. No claim that approval preceded all architecture is made.	B-01; Gate 1

Confirmed intake facts: team label, lecture source and team approval date. Open IDs retain the earlier numbering. Team approval is recorded; chronology compliance is not yet confirmed. [Q-12]

08 / GATE 1

Requirements approval

TEAM REQUIREMENTS APPROVAL
The team has read and approved the requirements F-01-F-21 and NF-01-NF-15 in this report, including the explicitly recorded deferrals. Approval date: 26 October 2026. Historical Gate 1 chronology remains open in Q-12.

Gate scope and historical chronology

B-01 requires Gate 1 before architecture is written. The team confirms 26 October 2026 as the approval date; existing architecture records are dated 24 October 2026. This report records the confirmed approval date and discloses that historical conflict rather than asserting the required sequence. Instructor clarification is needed for that issue. Architecture and detailed budgets belong to M3/Gate 2 under L-01 pp. 7-8. [NF-14; Q-12]

Submission check	Report coverage
Purpose, application and physical loop	Section 01; F-01-F-08, F-13, F-16-F-20.
F-/NF- requirements with reasons and acceptance	21 functional and 15 nonfunctional entries; sections 02-03.
Every timed requirement has units and lecture basis	Section 04; exact project magnitudes are distinguished from the lecture measurement method.
Every DRS and prerequisite embedded row considered	13 DRS and 8 embedded rows mapped in section 05; deferrals have reasons and role owners.
Decisions and open questions	Section 06 gives reasons and requirement traces; section 07 gives role owners and closure evidence.
Team name, milestone, date and Gate 1	MotionTennis; M2; report 11 November 2026; team approval 26 October 2026. Historical approval-before-architecture conflict remains OPEN: Q-12.
Anonymous, self-contained single PDF	No student identities, credentials, Git histories or repository-file dependencies. Required content and checklist reconciliation are reproduced.

Evidence and responsibility

The team remains responsible for requirement correctness and the final submission. AI assisted consolidation and formatting. No target experiment, measured performance, safety certification or hard worst-case guarantee is claimed by this requirements approval. [NF-04, NF-05, NF-09, NF-10, NF-14]